



# LOGS and EXPO BOS

## Part A: Revision of Indices

2 unit

TEACHER'S COPY

Q 1

Evaluate  $e^{-0.5}$  correct to three decimal places.

$$\underline{0.6065...=0.607(3dp)}$$

2006

Q 3

What is 0.005 233 59 written in scientific notation, correct to 4 significant figures?

2015

(A)  $5.2336 \times 10^{-2}$

(B)  $5.234 \times 10^{-2}$

(C)  $5.2336 \times 10^{-3}$

(D)  $5.234 \times 10^{-3}$

$$0.005\ 233\ 59 = 5.234 \times 10^{-3}$$

Answer: D

Q 2

Solve  $2^{2x+1} = 32$ .

$$2^{2x+1} = 32$$

$$2^{2x+1} = 2^5$$

$$2x + 1 = 5$$

$$x = 2$$

2011





# LOGS and EXPO BOS

## Part B: Logarithmic LAWS

2 unit

TEACHER'S COPY

**Q 1** Let  $\log_a 2 = x$  and  $\log_a 3 = y$ .

Find an expression for  $\log_a 12$  in terms of  $x$  and  $y$ .

$$\begin{aligned}\log_a 2 &= x, & \log_a 3 &= y \\ \log_a 12 &= \log_a 4 \times 3 \\ &= \log_a 4 + \log_a 3 \\ &= \log_a 2^2 + \log_a 3 \\ &= 2\log_a 2 + \log_a 3 \\ &= 2x + y.\end{aligned}$$

**Q 2** Let  $a = e^x$ .

Which expression is equal to  $\log_e(a^2)$ ?

- (A)  $e^{2x}$
- (B)  $e^{x^2}$
- (C)  $2x$
- (D)  $x^2$

1998

**Q 3** Which expression is equivalent to  $4 + \log_2 x$ ?

2016

- (A)  $\log_2(2x)$
- (B)  $\log_2(16 + x)$
- (C)  $4\log_2(2x)$
- (D)  $\log_2(16x)$

**Q 4** Write  $\log 2 + \log 4 + \log 8 + \dots + \log 512$  in the form  $a \log b$  where  $a$  and  $b$  are integers greater than 1.

2016

$$\begin{aligned}\log 2 + \log 4 + \log 8 + \dots + \log 512 \\ &= \log 2 + \log 2^2 + \log 2^3 + \dots + \log 2^9 \\ &= \log 2 + 2\log 2 + 3\log 2 + \dots + 9\log 2 \\ &= \frac{9}{2}(\log 2 + 9\log 2) \\ &= \frac{9}{2} \times 10\log 2 \\ &= 45\log 2\end{aligned}$$



# LOGS and EXPO BOS

## Part B: Logarithmic LAWS

2 unit

TEACHER'S COPY



Q 5

It is given that  $\ln a = \ln b - \ln c$ , where  $a, b, c > 0$ .

2017

Which statement is true?

A.  $a = b - c$

B.  $a = \frac{b}{c}$

C.  $\ln a = \frac{b}{c}$

D.  $\ln a = \frac{\ln b}{\ln c}$



# LOGS and EXPO BOS

## PART C: SOLVING EQUATIONS

**2 unit**

TEACHER'S COPY

**Q 1**

Solve  $\log_2(3x - 4) = 5$ .

$$\begin{aligned}\log_2(3x - 4) &= 5 \\ 3x - 4 &= 2^5 \\ 3x - 4 &= 32 \\ 3x &= 36 \\ \therefore x &= 12.\end{aligned}$$

**Q 2**

Solve the following equation for  $x$ :

$$2e^{2x} - e^x = 0.$$

$$2e^{2x} - e^x = 0$$

$$e^x(2e^x - 1) = 0$$

$\therefore e^x = 0$  no solution

$$\text{or, } 2e^x - 1 = 0$$

$$e^x = \frac{1}{2} \therefore x = \log_e \left( \frac{1}{2} \right)$$

**Q 3**

Solve  $\log_e x - \frac{3}{\log_e x} = 2$ .

$$\text{Q7a } \log_e x - \frac{3}{\log_e x} = 2, (\log_e x)^2 - 2\log_e x - 3 = 0,$$

$$(\log_e x - 3)(\log_e x + 1) = 0.$$

$$\therefore \log_e x = 3, x = e^3, \text{ or } \log_e x = -1, x = \frac{1}{e}.$$

2003

**Q 4**

What is the solution of  $5^x = 4$ ?

$$(A) \quad x = \frac{\log_e 4}{5}$$

$$(B) \quad x = \frac{4}{\log_e 5}$$

$$(C) \quad x = \frac{\log_e 4}{\log_e 5}$$

$$(D) \quad x = \log_e \left( \frac{4}{5} \right)$$

$$5^x = 4, \log_e 5^x = \log_e 4, x \log_e 5 = \log_e 4, x = \frac{\log_e 4}{\log_e 5}$$

C

**Q 5**

What is the solution to the equation  $\log_2(x - 1) = 8$ ?

$$(A) \quad 4$$

$$(B) \quad 17$$

$$(C) \quad 65$$

$$(D) \quad 257$$

$$\log_2(x - 1) = 8$$

$$\Rightarrow x - 1 = 2^8$$

$$\therefore x = 257$$

2007

2008

2013

2014



# LOGS and EXPO BOS

## Part D: graphing

2 unit

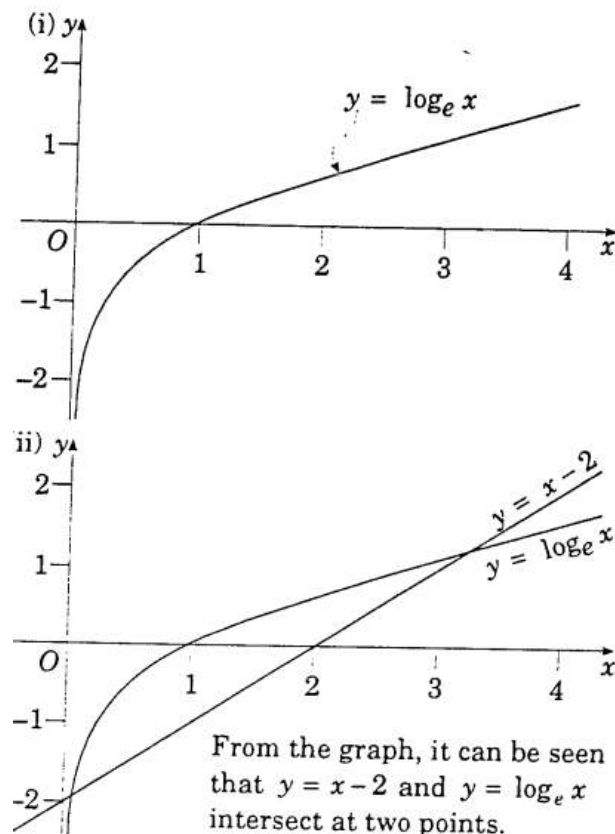
TEACHER'S COPY

Q 1

2000

- (i) Without using calculus, sketch  $y = \log_e x$ .
- (ii) On the same sketch, find, graphically, the number of solutions of the equation

$$\log_e x - x = -2.$$



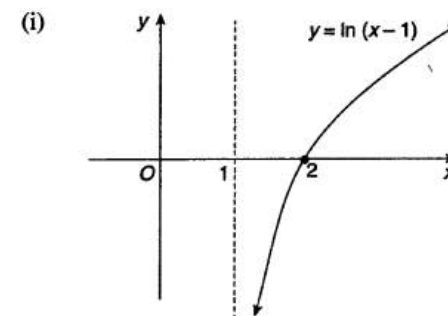
$\therefore$  There are two solutions to  $\log_e x = x - 2$ .  
That is, there are two solutions to  $\log x - x = -2$ .

Q 2

Consider the function  $y = \ln(x - 1)$  for  $x > 1$ .

2002

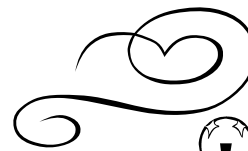
- (i) Sketch the function, showing its essential features.





# LOGS and EXPO BOS

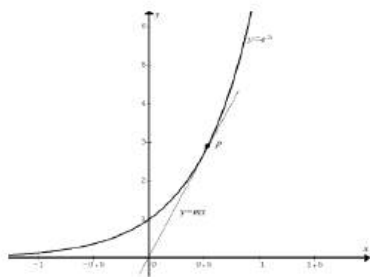
## Part D: graphing



**Q 3**

The line  $y = mx$  is a tangent to the curve  $y = e^{2x}$  at a point  $P$ .

- Sketch the line and the curve on one diagram.
- Find the coordinates of  $P$ .
- Find the value of  $m$ .



$$mx = e^{2x} \dots (1)$$

At  $P$ , the two functions have the same gradient:  $m = 2e^{2x} \dots (2)$

$$\therefore 2e^{2x} \cdot x = e^{2x}$$

$$e^{2x} (2x - 1) = 0$$

$$\therefore x = \frac{1}{2}$$

$$\therefore P\left(\frac{1}{2}, e\right)$$

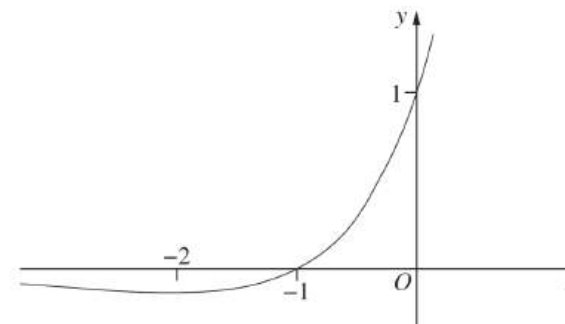
$$\text{At } P, e^{2x} = mx, \therefore e = \frac{m}{2}, m = 2e$$

**2014**

**Q 4**

The diagram shows the graph of  $y = e^x(1+x)$ .

**2015**



How many solutions are there to the equation  $e^x(1+x) = 1 - x^2$ ?

- 0
- 1
- 2**
- 3

Draw the parabola (with intercepts  $(0,1)$ ,  $(-1,0)$ ,  $(1,0)$ ) over the graph. There are two intersections.

Answer: C

Q 1

Find the equation of the tangent to the curve

$$y = 3x + e^{-x}$$

1987

at the point where  $x = 0$ .

i)  $y = 3x + e^{-x}$   
at  $x = 0$ ,  $y = 3(0) + e^{-0}$   
 $= 0 + 1$   
 $= 1$

$\therefore$  Point is  $(0, 1)$

$$\frac{dy}{dx} = 3 - e^{-x}$$

at  $x = 0$ ,  $\frac{dy}{dx} = 3 - e^{-0}$   
 $= 3 - 1$   
 $= 2$

Equation of tangent at  $(0, 1)$  with gradient  $= 2$  is given by

$$y - y_1 = m(x - x_1)$$

i.e.  $y - 1 = 2(x - 0)$

$$y - 1 = 2x$$

$$y = 2x + 1.$$

Q 2

Differentiate:  $xe^{2x}$

1988

$$\frac{d}{dx}(xe^{2x})$$

$$= x \cdot \frac{d}{dx}(e^{2x}) + e^{2x} \cdot \frac{d}{dx}(x)$$

$$= x \cdot 2e^{2x} + e^{2x} \cdot 1$$

$$= 2xe^{2x} + e^{2x}$$

$$= e^{2x}(2x + 1).$$

Q 3

Differentiate:  $e^{3x}$

1991

$$\frac{d}{dx}(e^{3x}) = e^{3x} \cdot \frac{d}{dx}(3x)$$

$$= e^{3x} \cdot 3$$

$$= 3e^{3x}$$

Q 4

Differentiate:  $4e^{2x}$

1995

$$\begin{aligned}\frac{d}{dx}(4e^{2x}) &= 4e^{2x} \cdot \frac{d}{dx}(2x) \\ &= 4e^{2x} \cdot 2 \\ &= 8e^{2x}.\end{aligned}$$

Q 5

Differentiate:  $xe^{2x}$

1996

$$\begin{aligned}\frac{d}{dx}(xe^{2x}) &= x \cdot \frac{d}{dx}(e^{2x}) + e^{2x} \cdot \frac{d}{dx}(x) \\ &= x \cdot 2e^{2x} + e^{2x} \cdot 1 \\ &= 2xe^{2x} + e^{2x}.\end{aligned}$$

Q 6

Differentiate  $\frac{e^x}{1+x}$ .

1997

$$\begin{aligned}\frac{d}{dx}\left(\frac{e^x}{1+x}\right) &= \frac{(1+x) \times e^x - e^x \times 1}{(1+x)^2} \\ &= \frac{e^x + xe^x - e^x}{(1+x)^2} \\ &= \frac{xe^x}{(1+x)^2}.\end{aligned}$$

Q 7

Differentiate  $\frac{x}{e^x}$ .

2001

$$\begin{aligned}\frac{d}{dx}\left(\frac{x}{e^x}\right) &= \frac{e^x \cdot 1 - xe^x}{(e^x)^2} \quad (\text{using quotient rule}) \\ &= \frac{e^x(1-x)}{e^{2x}} \\ &= \frac{1-x}{e^x}.\end{aligned}$$



Q 8

Find the equation of the tangent to  $y = e^{2x}$  at the point  $(0, 1)$ .

2002

$$y = e^{2x}$$

$$\frac{dy}{dx} = 2e^{2x}$$

at  $x = 0$ ,  $\frac{dy}{dx} = 2e^0$   
 $= 2$

$\therefore$  Equation of tangent at  $(0, 1)$  with gradient 2

$$\text{is } y - 1 = 2(x - 0)$$

$$y - 1 = 2x$$

$$y = 2x + 1.$$

Q 9

Differentiate  $(2e^x - 4)^9$

2002

$$\frac{d}{dx} [(2e^x - 4)^9]$$

$$= 9(2e^x - 4)^8 \cdot \frac{d}{dx} (2e^x - 4)$$

$$= 9(2e^x - 4)^8 \cdot (2e^x)$$

$$= 18e^x(2e^x - 4)^8.$$

Q 10

Find the coordinates of the point  $P$  on the curve  $y = 2e^x + 3x$  at which the tangent to the curve is parallel to the line  $y = 5x - 3$ .

2005

$$y = 2e^x + 3x$$

$$\frac{dy}{dx} = 2e^x + 3.$$

Since the tangent is parallel to  $y = 5x - 3$ ,  
then gradient of the tangent is 5.

ie.  $\frac{dy}{dx} = 5$

$$2e^x + 3 = 5$$

$$2e^x = 2$$

$$e^x = 1$$

$$\therefore x = 0.$$

When  $x = 0$ ,  $y = 2e^0 + 3(0)$   
 $= 2.$

$\therefore$  The coordinates are  $P(0, 2)$ .

Q 11 Find the derivative of  $y = x^2 e^x$  with respect to  $x$ .

2011

$$\begin{aligned}\frac{dy}{dx} &= x^2 \cdot e^x + e^x \cdot 2x \\ &= e^x (x^2 + 2x)\end{aligned}$$

Q 12 Differentiate  $(3 + e^{2x})^5$

2012

$$\frac{d}{dx} (3 + e^{2x})^5 = 5(3 + e^{2x})^4 (2e^{2x}) = 10e^{2x} (3 + e^{2x})^4$$

Q 13 Differentiate  $(e^x + x)^5$

2015

$$\frac{d}{dx} (e^x + x)^5 = 5(e^x + 1)(e^x + x)^4$$

Q 14 Differentiate  $y = xe^{3x}$ .

2016

$$\begin{aligned}y &= xe^{3x} \\ \frac{dy}{dx} &= 3xe^{3x} + e^{3x} \\ &= e^{3x} (1 + 3x)\end{aligned}$$

Q 15 What is the derivative of  $e^{x^2}$ ?

2017

- A.  $x^2 e^{x^2-1}$
- B.  $2xe^{2x}$
- C.  $2xe^{x^2}$
- D.  $2e^{x^2}$

Q 16 Differentiate  $\frac{e^x}{x+1}$ .

2018

$$\begin{aligned}\frac{d}{dx} \left( \frac{e^x}{x+1} \right) &= \frac{(x+1) \times e^x - e^x \times 1}{(x+1)^2} \\ &= \frac{xe^x + e^x - e^x}{(x+1)^2} \\ &= \frac{xe^x}{(x+1)^2}\end{aligned}$$



# LOGS and EXPO BOS

## Part F: General Application

**2 unit**

**TEACHER'S COPY**

**Q 1**

**2008**

Light intensity is measured in lux. The light intensity at the surface of a lake is 6000 lux. The light intensity,  $I$  lux, a distance  $s$  metres below the surface of the lake is given by

$$I = Ae^{-ks}$$

where  $A$  and  $k$  are constants.

- (i) Write down the value of  $A$ . **1**
- (ii) The light intensity 6 metres below the surface of the lake is 1000 lux. Find the value of  $k$ . **2**
- (iii) At what rate, in lux per metre, is the light intensity decreasing 6 metres below the surface of the lake? **2**

Q5c  $I = Ae^{-ks}$  and  $I = 6000$  when  $s = 0$ .  $\therefore A = 6000$ .

Q5cii  $I = 6000e^{-ks}$  and  $I = 1000$  when  $s = 6$ .

$$\therefore 1000 = 6000e^{-6k}, \quad e^{-6k} = \frac{1}{6}, \quad e^{6k} = 6, \quad 6k = \log_e 6,$$

$$\therefore k = \frac{1}{6} \log_e 6 \approx 0.2986.$$

Q5ciii  $I = Ae^{-ks}, \quad \frac{dI}{ds} = -kAe^{-ks} = -kI.$

When  $s = 6, I = 1000,$

$$\frac{dI}{ds} = -kI = -\left(\frac{1}{6} \log_e 6\right) 1000 = -\frac{500}{3} \log_e 6 \approx -298.6$$

The light intensity is decreasing at 298.6 lux per metre at 6 metres below the surface of the lake.

**Q 2**

**2011**

The intensity  $I$ , measured in watt/m<sup>2</sup>, of a sound is given by

$$I = 10^{-12} \times e^{0.1L},$$

where  $L$  is the loudness of the sound in decibels.

- (i) If the loudness of a sound at a concert is 110 decibels, find the intensity of the sound. Give your answer in scientific notation. **1**
- (ii) Ear damage occurs if the intensity of a sound is greater than  $8.1 \times 10^{-9}$  watt/m<sup>2</sup>. **2**  
What is the maximum loudness of a sound so that no ear damage occurs?
- (iii) By how much will the loudness of a sound have increased if its intensity has doubled? **2**

Consider two intensities:

$$I = 10^{-12} e^{\frac{110}{10}} = 10^{-12} e^{11} \\ \approx 5.99 \times 10^{-8}$$

$$I_1 = 10^{-12} e^{\frac{L_1}{10}}, \quad I_2 = 10^{-12} e^{\frac{L_2}{10}}$$

and assume  $I_2 = 2I_1$ . Hence,

$$8.1 \times 10^{-9} = 10^{-12} e^{\frac{110}{10} + 0.1L}$$

$$I_2 = 10^{-12} e^{\frac{L_2}{10}} = 2 \times 10^{-12} e^{\frac{L_1}{10}} = 2I_1$$

$$8.1 \times 10^3 = e^{\frac{L}{10}}$$

$$e^{\frac{L_2}{10}} = 2e^{\frac{L_1}{10}}$$

$$\ln 8100 = \frac{L}{10}$$

$$\frac{L_2}{10} = \ln \left( 2e^{\frac{L_1}{10}} \right) = \ln 2 + \frac{L_1}{10}$$

$$L = 10 \ln 8100 \approx 89.996$$

$$L_2 - L_1 = 10 \ln 2$$

$$= 90 \text{ decibels}$$

